

Lecture 07

Probability, Statistics, Uncertainty, and Stochastic Simulations

Part II

Remaining Future Schedule (~End of Course)

❖ ~~May-17 (1): TEST~~

❖ **May-24 (Today's Lecture):**

- ◆ **Introduction to Final Project B**

- ◆ **Lec 07: Probability, Statistics, Uncertainty, and Stochastic Simulations II (including Lec 08: Performance Criteria)**

❖ **May-31 & Jun-07 (3 & 4)**

- ◆ **Lec 09: Water Resources Systems Engineering Frontiers**
-Introduction and Examples Part 1

- ◆ **Lec 10: Water Resources Systems Engineering Frontiers**
-Introduction and Examples Part 2 (*Invited Talk*)

❖ **Jun-14: Final Project Presentation + Report Due (Jun-20)**

- ◆ Report는 추후에 **기술기사**로 출판할 예정 (7월 중): **기술기사 예시**

Final Project B

❖ 이제, 이 수업에서 배운 내용을 활용하여 보령댐의 최적 운영방법을 도출해 봅시다. 알다시피, 보령댐은 특히 최근 다년 가뭄으로 인한 피해를 많이 입은 바 있습니다. 팀별로 아래 문제에 대한 답변을 PPT 발표에 포함하여 준비하여 주시기 바랍니다. (15분 발표, 5분 Q&A)

(팀 C: 이재황 김현주 박성열 // 팀 D: 이재호 김연주 남현욱 이주형)

- ◆ 보령댐 및 보령댐 유역에 대한 예비 분석을 수행합니다. 직면하고 있는 문제는 무엇이며, 직면한 문제의 어떤 부분을 개선할 수 있습니까?
- ◆ 보령댐이 최근 겪은 가뭄은 얼마나 심각했습니까? 이벤트에 대한 확률 분석을 수행하여 봅시다.
- ◆ 대상 댐에 적용할 최적화 모델의 종류를 선택하고, 문제를 설정해 봅시다. 최적화를 위한 목적 함수, 결정 변수, 제약 조건은 각각 무엇인가요? 결정적/확률적 모델 중 어떤 모델을 사용해야 하나요? (R을 통한 Coding 필수)
- ◆ 최적화된 운영률을 가지고 2014년 9월 1일부터 2019년 6월 30일까지의 일별 저수지 모의를 진행해 봅시다. (초기 수위는 2014년 8월 30일 저수량)
- ◆ 최적화 결과 및 모의 결과를 사용하여 분석을 진행해 봅시다. 실제 운영 실적과 비교하여 최적 운영률이 어떤 측면에서 얼마나 개선된 결과를 보였나요?

Final Project B 최종 리포트 목차

이전에 설명드린 바와 같이, 조별로 제출한 최종 리포트는 학기가 끝난 후 정리 및 취합 과정을 거쳐 7월 중 학술기사(한국수자원학회지-물과미래)에 투고할 예정입니다. 예시로 제시한 학술 기사를 참고해 주시고, 한글로 리포트 작성 부탁드립니다. 통일성을 위해, 되도록이면 첨부 파일 양식에 아래와 같은 목차를 사용하여 작성 부탁드립니다 (리포트 제출 기한: Jun-20 오후 6시까지 메일로 제출).

- I. 서론 (들어가며)
- II. 개요 (대략적인 최적화 모형 설명 및 선택한 최적화 모형 이유)
- III. 대상 유역 및 문제 제기
- IV. 문제 설정 및 최적화 진행 과정 (최적화 모델 세부 구성 및 설명 추가 필요)
- V. 최적화 진행 결과 및 과거 유입량 바탕의 모의 진행 결과
- VI. 결론 및 토론 (+시사점&한계점?)

R) Finally, Conditional Probability $P(A|B)$

4. Conditional probability

Probability of occurrence of event **A** given that event **B** has already occurred is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

or

$$P(A \cap B) = P(B)P(A|B) = P(A)P(B|A)$$

If $P(A|B) = P(A)$ then the events are said to be **statistically independent**

Example:

For daily rainfall in Ashland, Kentucky it is known that the conditional probability for a wet day (W) following a dry day (D), $P(W|D) = 0.444$ (this is also called a transition probability).

Similarly:

- $P(W|D) = 0.276$
- $P(D|D) = 0.724$
- $P(D|W) = 0.556$

What is the probability that a wet day is followed by two more wet days?

Let A be the event that day 2 is wet and B be the event that day 3 is wet. Then

$$P(A \cap B) = P(W|W)P(W|W) = 0.444^2 = 0.197$$

R) Bayes Theorem

❖ The basics

◆ *a priori* uncertainty + new information

- Yields revised assessment of uncertainty
- Multiple observations—use repetitions of Bayes Theorem

Bayes Theorem

$$P(E|O) = P(E) \left\{ \frac{P(O|E)}{P(O)} \right\}$$

$P(E)$ = Prior probability of event E

$P(E|O)$ = Posterior probability of event E given observation O

$P(O|E)$ = conditional probability of the observation O and the event E

$P(O)$ = Probability of making observation O

❖ $P(E/O)$

- ◆ Read “probability of event E conditional on O having been observed”
- ◆ New information in estimate

❖ $P(O/E)$

- ◆ Frequency that observation O is associated with existence of specific event E
- ◆ Ex: probability of observing salt domes when there is oil

❖ $P(E/O) \neq P(O/E)$ these probabilities are not symmetric!

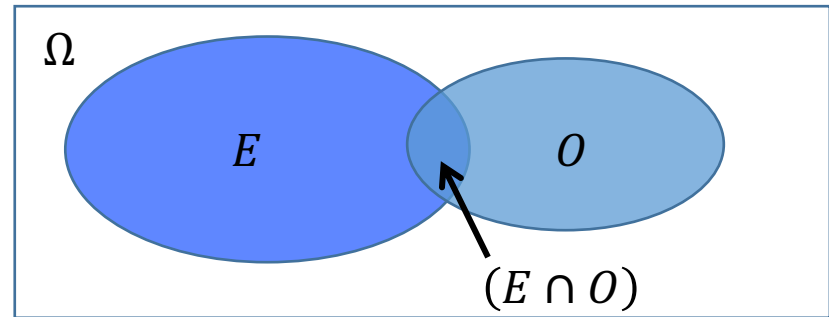
- ◆ $P(O = \text{male} / E = \text{king}) = 1.0$
- ◆ $P(E = \text{king} / O = \text{male})$ is probability of finding a king given the observation an individual is male = $\sim 10^{-9}$

❖ $P(O)$

- ◆ Probability of making a specific observation O considering all ways it can occur
- ◆ The probability of making observation O may be associated with all possible events E_i

$$P(O) = \sum_i P(O | E_i) P(E_i)$$

R) Bayes Theorem: Derivation



According to conditional probability,

$$P(E|O) = \frac{P(E \cap O)}{P(O)} \quad \text{The fraction of ellipse B that overlaps with ellipse A}$$

and

$$P(O|E) = \frac{P(E \cap O)}{P(E)} \quad \text{The fraction of ellipse A that overlaps with ellipse B.}$$

Solving for $P(A \cap B)$, the common factor:

$$P(E \cap O) = P(E)P(O|E) = P(O)P(E|O)$$

Rearranging to give $P(A|B)$:

$$P(E|O) = \frac{P(E)P(O|E)}{P(O)} \quad \text{Note: one can reverse E and O and the solution is still valid}$$

Example Problem 1: Bayes Theorem

EXAMPLE 2.30 An individual has 3 different email accounts. Most of her messages, in fact 70%, come into account #1, whereas 20% come into account #2 and the remaining 10% into account #3. Of the messages into account #1, only 1% are spam, whereas the corresponding percentages for accounts #2 and #3 are 2% and 5%, respectively. What is the probability that a randomly selected message is spam?

To answer this question, let's first establish some notation:

$A_i = \{\text{message is from account \# } i\}$ for $i = 1, 2, 3$, $B = \{\text{message is spam}\}$

Then the given percentages imply that

$$P(A_1) = .70, P(A_2) = .20, P(A_3) = .10$$

$$P(B|A_1) = .01, P(B|A_2) = .02, P(B|A_3) = .05$$

Now it is simply a matter of substituting into the equation for the law of total probability:

$$P(B) = (.01)(.70) + (.02)(.20) + (.05)(.10) = .016$$

In the long run, 1.6% of this individual's messages will be spam. ■

Example Problem 2: Bayes Theorem

EXAMPLE 2.31 *Incidence of a rare disease.* Only 1 in 1000 adults is afflicted with a rare disease for which a diagnostic test has been developed. The test is such that when an individual actually has the disease, a positive result will occur 99% of the time, whereas an individual without the disease will show a positive test result only 2% of the time (the *sensitivity* of this test is 99% and the *specificity* is 98%; in contrast, the [Sept. 22, 2012 issue of *The Lancet*](#) reports that the first at-home HIV test has a sensitivity of only 92% and a specificity of 99.98%). If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?

To use Bayes' theorem, let A_1 = individual has the disease, A_2 = individual does not have the disease, and B = positive test result. Then $P(A_1) = .001$, $P(A_2) = .999$, $P(B|A_1) = .99$, and $P(B|A_2) = .02$. The tree diagram for this problem is in Figure 2.12.

Example Problem 2: Bayes Theorem

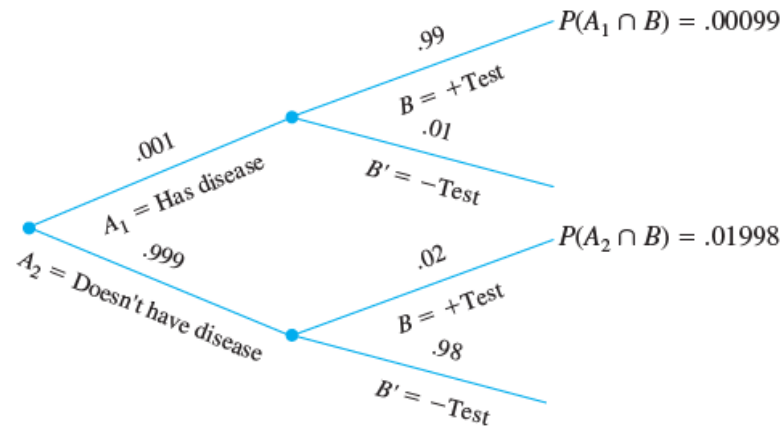


Figure 2.12 Tree diagram for the rare-disease problem

Next to each branch corresponding to a positive test result, the multiplication rule yields the recorded probabilities. Therefore, $P(B) = .00099 + .01998 = .02097$, from which we have

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{.00099}{.02097} = .047$$

This result seems counterintuitive; the diagnostic test appears so accurate that we expect someone with a positive test result to be highly likely to have the disease, whereas the computed conditional probability is only .047. However, the rarity of the disease implies that most positive test results arise from errors rather than from diseased individuals. The probability of having the disease has increased by a multiplicative factor of 47 (from prior .001 to posterior .047); but to get a further increase in the posterior probability, a diagnostic test with much smaller error rates is needed. ■

Today's Topic: Probability, Statistics, Uncertainty and Stochastic Simulations

❖ Part I: Last Class (May-10)

- ◆ Review on Common Representative Values
- ◆ Concept of Probability & Review on the Probability Concepts
- ◆ Common Distributions in WRSA: 68-95-99.7 rule for Norm. Dist.
- ◆ Conditional Probability & Bayes Theorem

❖ Part II: This Class (May-24)

- ◆ Details & Procedures for Probabilistic Modeling
- ◆ Methods for Synthetic Streamflow Generations
- ◆ Performance Criteria

Probabilistic Modeling

❖ Steps for Probabilistic Modeling

- ◆ Pick a Model: Uniform, Normal (or Lognormal), Exponential, etc.
- ◆ Fit the model to Data: MLE
 - Maximum Likelihood Estimator (MLE)

Let $X_1, X_2, \dots, X_n$ have joint pmf or pdf

$$f(x_1, x_2, \dots, x_n; \theta_1, \dots, \theta_m) \quad (6.6)$$

where the parameters $\theta_1, \dots, \theta_m$ have unknown values. When $x_1, \dots, x_n$ are the observed sample values and (6.6) is regarded as a function of $\theta_1, \dots, \theta_m$, it is called the **likelihood function**. The maximum likelihood estimates (mle's) $\hat{\theta}_1, \dots, \hat{\theta}_m$ are those values of the θ_i 's that maximize the likelihood function, so that

$$f(x_1, \dots, x_n; \hat{\theta}_1, \dots, \hat{\theta}_m) \geq f(x_1, \dots, x_n; \theta_1, \dots, \theta_m) \text{ for all } \theta_1, \dots, \theta_m$$

When the X_i 's are substituted in place of the x_i 's, the **maximum likelihood estimators** result.

❖ How to pick a model?

- ◆ Physical basis: model designed for properties of physical process
- ◆ Limit Laws
- ◆ Empirical Basis: which model fits data best?

Details on Probabilistic Modeling

❖ Then, how to pick a model?

◆ Physical basis: model designed for properties of physical process

Ex) Poisson distribution:

Number of time event occurs in an interval of time

* events are independent

* rate of occurrences is constant

* number of extreme events per year (especially in hydrology)

A discrete random variable X is said to have a **Poisson distribution** with parameter μ ($\mu > 0$) if the pmf of X is

$$p(x; \mu) = \frac{e^{-\mu} \cdot \mu^x}{x!} \quad x = 0, 1, 2, 3, \dots$$

Details on Probabilistic Modeling

❖ Then, how to pick a model?

◆ Limit Laws

- Theory suggest a distribution as $n \rightarrow \infty$

Ex) Central Limit Theorem (CLT)

The Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and variance σ^2 . Then if n is sufficiently large, $\bar{X}$ has approximately a normal distribution with $\mu_{\bar{X}} = \mu$ and $\sigma_{\bar{X}}^2 = \sigma^2/n$, and T_o also has approximately a normal distribution with $\mu_{T_o} = n\mu$, $\sigma_{T_o}^2 = n\sigma^2$. The larger the value of n , the better the approximation.

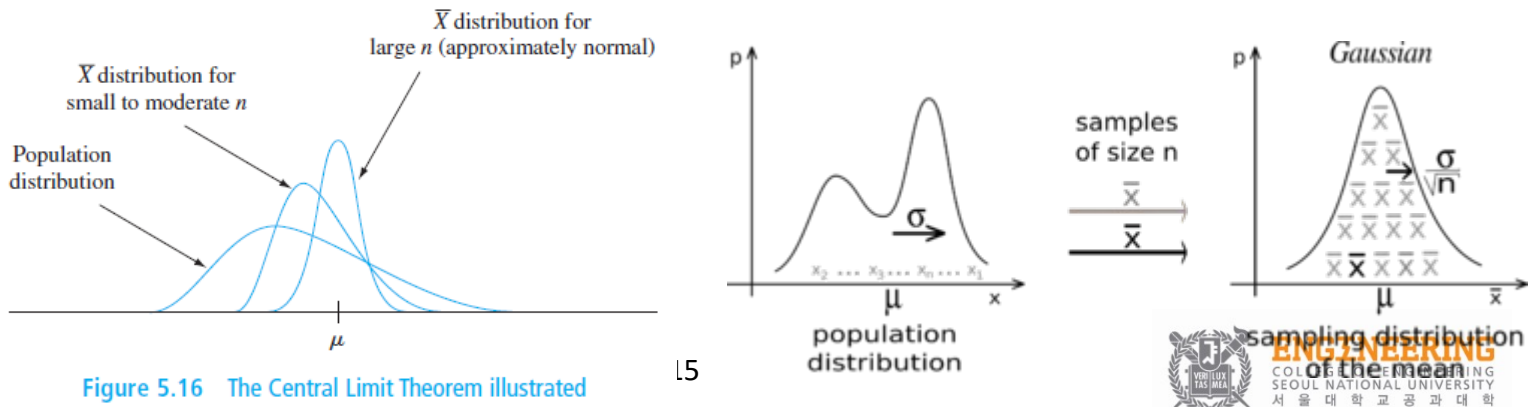


Figure 5.16 The Central Limit Theorem illustrated

Details on Probabilistic Modeling

❖ Then, how to pick a model?

◆ Empirical Basis: which model fits data best?

* Goodness-of-fit Approaches such as:

- 1) Picking model based on scores (AIC)
- 2) Visual Assessments (P-P or Q-Q plots)
- 3) Hypothesis testing (Chi-squared test)

* AIC, Akaike Information Criterion (아카이케 정보 기준)

$$AIC = -2 \ln \theta_{MLE} + 2k$$

in which the goal is to pick the model with the smallest AIC
(=large maximum likelihood, penalty for model parameter)

Details on Probabilistic Modeling

❖ Then, how to pick a model?

◆ Empirical Basis: which model fits data best?

* AIC, Akaike Information Criterion (아카이케 정보 기준)

$$AIC = -2 \ln \theta_{MLE} + 2k$$

in which the goal is to pick the model with the smallest AIC
(=large maximum likelihood, penalty for model parameter)

Ex) Exp:

Gamma:

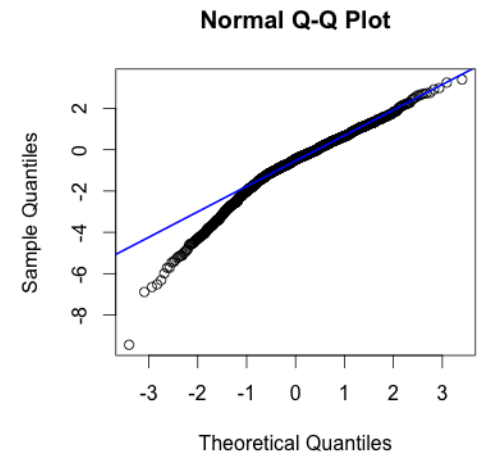
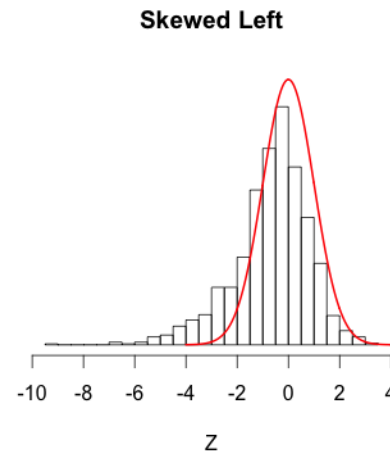
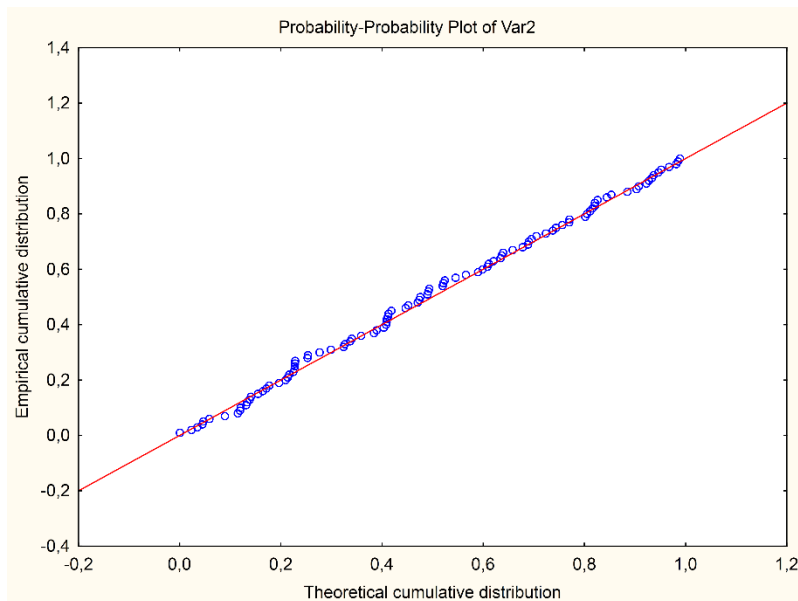
* Rule of Thumb *

Details on Probabilistic Modeling

❖ Then, how to pick a model?

◆ **Empirical Basis: which model fits data best?**

2) Visual Assessments (P-P or Q-Q plots)

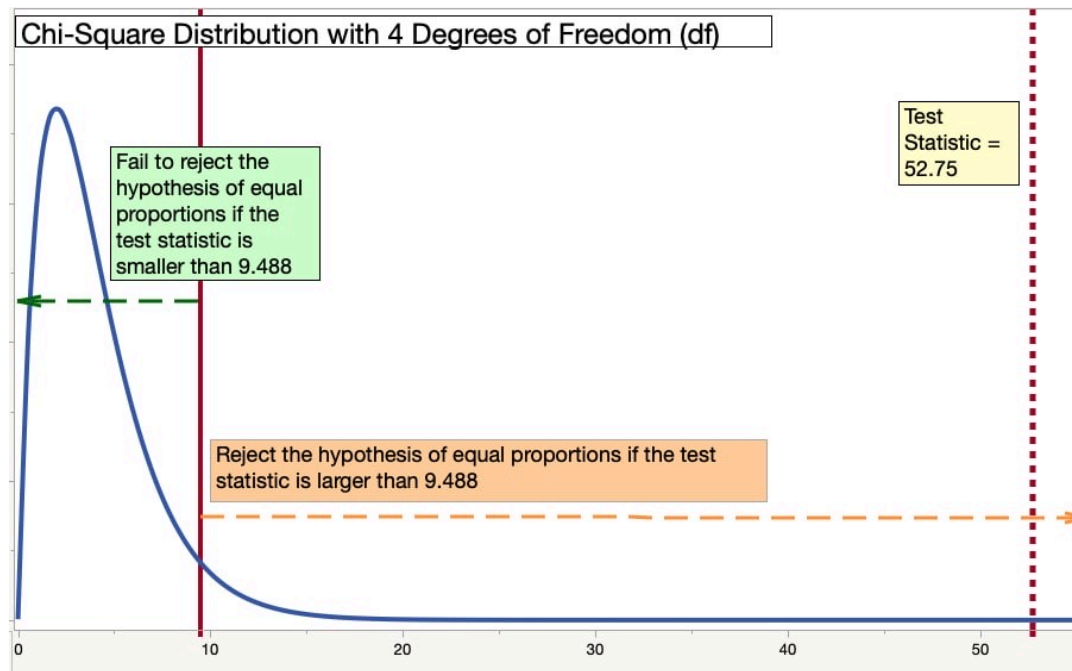


Details on Probabilistic Modeling

❖ Then, how to pick a model?

◆ **Empirical Basis: which model fits data best?**

3) Hypothesis testing (Chi-squared test)



❖ For Further Details

◆ Refer to
2022 Fall Semester
Statistical Hydrology
(457.568A)

2021 Fall Semester
Statistical Hydrology
Syllabus

Week	Date	Topic	Assignment
2	9/6	Course Introduction ((P1)) Chap 2 Probability	
2	9/8	((P1)) Chap 3 Properties of Random Variables ((P1)) Chap 3 Parameter Estimation	
3	9/15	((P1)) Chap 4 Discrete Probability Distributions ((P1)) Chap 5 Normal Distribution Family	
4	9/22	추석	
5	9/29	((P1/P3)) Chap 6 GEV & Pearson Distribution Families ((P1/P3)) Chap 7 Flood Frequency Analysis	<i>Term Project I</i> <i>10-min Video Due</i>
6	10/6	((P4)) L-Moments	Invited Talk (성장현 박사)
7	10/13	((P4)) Regional FFA: Index Flood Method	Invited Talk (성장현 박사)
8	10/20	((S1/S2)) Regional Regression	Invited Talk (서승범 교수)
9	10/27	((P5/P6)) Statistical Properties of Hydrologic Time Series ((P5/P6)) Testing and Removing Nonstationarities	
10	11/3	Test	
11	11/10	((P5/P6)) AR Modeling: Identification	
12	11/17	((P5/P6)) AR: Parameter Est & Diagnostic Check	
13	11/24	((P5/P6)) Univariate ARMA ((P5/P6)) Periodic ARMA	
14	12/1	((P7)) Seasonal ARIMA ((P3/P5)) Multivariate Time Series Modeling	
15	12/8	((S3)) An Introduction to Ensemble Streamflow Prediction	Invited Talk (이재경 교수)
16	12/15	<i>Term Project II</i>	Presentation

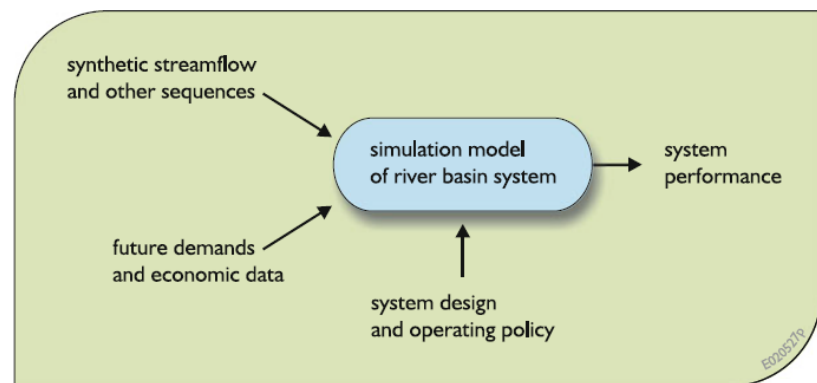
6.8 Synthetic Streamflow Generation (p.257)

❖ Why do we Generate Synthetic Streamflow?

- ◆ Little or no data for your system
- ◆ Evaluate the effect of hydrologic non-stationarity on your system

❖ Types of Synthetic streamflow generators

- ◆ Indirect approach: generate precipitation and temperature series and input them to a hydrologic model of a basin (GCM)
- ◆ **Direct approach**: uses statistical techniques to generate streamflow time series directly



6.8 Synthetic Streamflow Generation (p.257)

❖ Types of Synthetic streamflow generators

- ◆ **Direct approach**: uses statistical techniques to generate streamflow time series directly

❖ Types of direct approaches

◆ Parametric methods

- Relies on hypothesized statistical model of streamflow whose parameters are selected to achieve a desired result
- Time series model! (Autoregressive Moving Average- ARMA)

◆ Non-parametric methods

- Re-sample flows from a record & rely heavily on the data
- Examples include Block bootstrapping, k-nearest neighbor, etc.

Performance Criteria

❖ Reliability, Resilience & Vulnerability

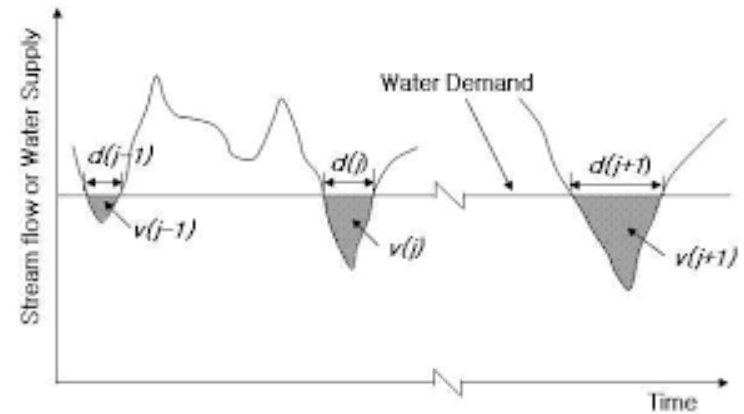


Table 3. Classification of selected performance indices

Number	Performance Indices	Definition	Classification
1	Average Deficit Frequency	$\beta(\%) = \frac{1}{T} \left(\sum_{t=1}^T X_t \in F \right) \times 100(\%)$	Frequency
2	Average Recovery Time	$\gamma(days) = \frac{1-\alpha}{\rho} = \frac{\left(\sum_{t=1}^T X_t \in F \right)}{\left(\sum_{t=1}^{T-1} X_t \in S, X_{t+1} \in F \right)}$	Length
3	Average Deficit Magnitude	$\omega(\%) = \frac{\left(\sum_{t=1}^T \max(0, D_t - R_t) \right)}{\left(\sum_{t=1}^T D_t \right)} \times 100(\%)$	Magnitude

Performance Criteria: Robustness

- ❖ Among multiple types of robustness measures,
 - ◆ **Regret-based:** cost of choosing incorrectly
(focuses on the incorrect choice of decision alternative)
(Savage 1951)
 - ◆ **Satisficing-based:** tendency of decision makers to seek outcomes that meet one or more requirements but not achieve optimal performance (Simon 1959)

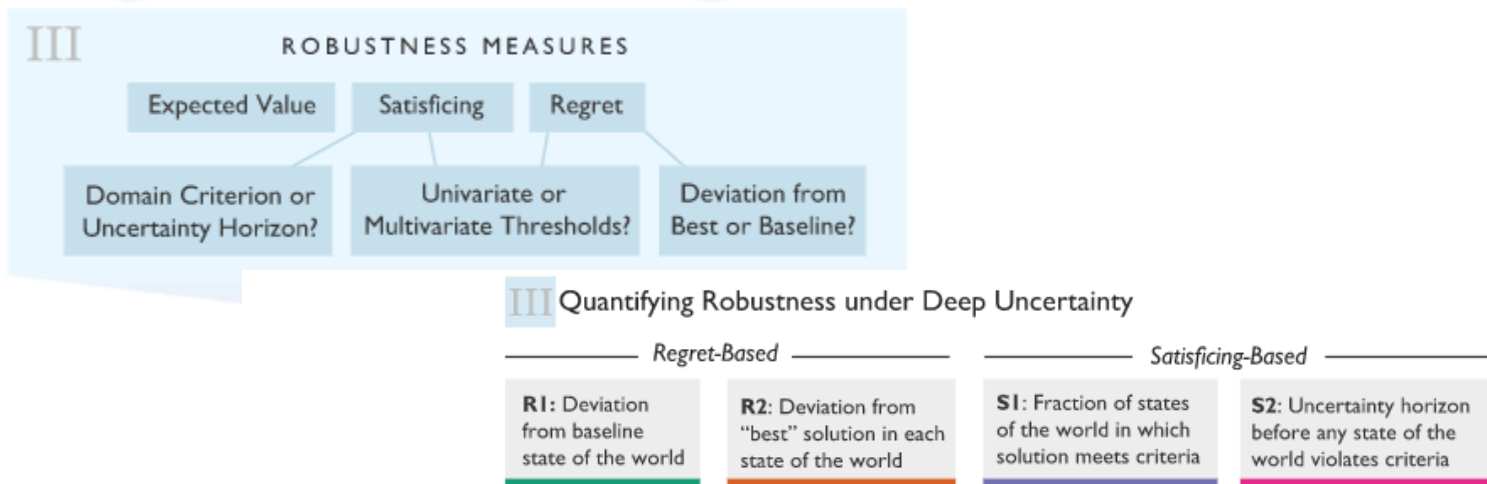


Fig. 3. Four measures for quantifying robustness; the categories of regret and satisficing categories are adapted from Lempert and Collins (2007); measure R2 is a variant of Savage regret (Savage 1951), and S1 links to Starr's domain criterion (Starr 1962; Schneller and Sphicas 1983); the uncertainty horizon, S2, is drawn from the Info-Gap method (Ben-Haim 2004)

Performance Criteria: Robustness

Here, we define R1 for each decision alternative as the 90th percentile deviation from the baseline, maximized over all objectives

$$R1 = \max_i [D_{i,90}:P(D_i \leq D_{i,90}) = 0.90] \quad (3)$$

$$D_{i,j} = \frac{|F(\mathbf{x})_{i,j} - F(\mathbf{x})_i^*|}{F(\mathbf{x})_i^*} \quad (4)$$

where $F(\mathbf{x})_i^*$ represents the value of objective i in the baseline state of the world; and $F(\mathbf{x})_{i,j}$ represents the value of objective i calculated in state of the world j .

Similarly, measure R2 calculates regret as the deviation from the best solution in each state of the world

$$R2 = \max_i [D_{i,90}:P(D_i \leq D_{i,90}) = 0.90] \quad (5)$$

$$D_{i,j} = \frac{|F(\mathbf{x})_{i,j} - \min_s F(\mathbf{x}_s)_{i,j}|}{F(\mathbf{x})_{i,j}} \quad (6)$$

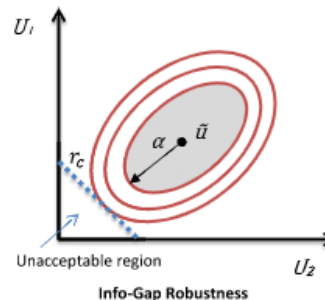
Measure S1 is defined as the fraction of N states of the world in which a solution meets stakeholders' performance requirements in one or more objectives

$$S1 = \frac{1}{N} \sum_{j=1}^N \Lambda_{s,j} \quad (7)$$

where $\Lambda_{s,j} = 1$ if solution s meets requirements in state of the world j and $\Lambda_{s,j} = 0$ otherwise. An important component of this measure is the elicitation of requirements from stakeholders. In this case study, the utilities indicated the following requirements: reliability above 99%, restriction frequency less than 20% (once every 5 years), and worst-case cost less than 5% of annual volumetric revenue (Herman et al. 2014). Controlling the worst-case cost is critical for utilities operating under the cost-recovery budget model, in which stable revenues are required to recoup large infrastructure investments (Hughes and Leurig 2013). The use of multivariate requirements has been advocated in the MORDM framework (Kasprzyk et al. 2013) and elsewhere (e.g., Gerst et al. 2013).

Finally, measure S2 represents the uncertainty horizon that can be withstood prior to system failure. Following the Info-Gap approach, the uncertainty horizon (or robustness) can be defined as (Hall et al. 2012)

$$\ell \quad S2 = \hat{\alpha} = \max \left[\alpha: \min_{j \in U(\alpha)} F(\mathbf{x})_j \geq r^* \right] \quad (8)$$



End of Lecture

❖ May-24 (Today's Lecture):

- ◆ Introduction to Final Project B
- ◆ Lec 07: Probability, Statistics, Uncertainty, and Stochastic Simulations II (including Lec 08: Performance Criteria)

❖ May-31 & Jun-07 (3 & 4)

- ◆ Lec 09: Water Resources Systems Engineering Frontiers
-Introduction and Examples Part 1
- ◆ Lec 10: Water Resources Systems Engineering Frontiers
-Introduction and Examples Part 2 (*Invited Talk*)